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Comparison of virus detection, productivity, and economic performance between lots of growing pigs vaccinated with two doses or one dose of PRRS MLV vaccine, under field conditions

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ABSTRACT

Porcine reproductive and respiratory syndrome virus (PRRSV) causes a significant economic impact on swine production. It has been demonstrated that PRRS modified-live virus (MLV) vaccination of pigs, with one full dose, significantly reduces clinical consequences of wild-type PRRSV infection compared to non-vaccinates. However, there is limited information about the effect that two doses of PRRSV MLV vaccine have on the performance of growing pigs, compared to vaccination with a single dose. This study was conducted with the objectives to compare (a) the wild-type PRRSV detection in oral fluids over time, (b) key closeout productivity indicators, and (c) economic performance between lots of growing pigs vaccinated with two doses of Ingelvac PRRS® MLV vaccine and lots vaccinated with a single dose of the same vaccine. This randomized field trial included 15 lots of growing pigs from PRRSV positive-unstable sow farms and 66 lots from PRRSV positive-stable sow farms, according to the American association of swine veterinarians' terminology. All pig lots received the first vaccination either around processing or weaning age. Lots allocated in the two doses group received the second vaccination three to four weeks after the first vaccination. The pig lots were monitored for PRRSV detection over time. Six oral fluids samples were collected in three weeks intervals and were tested for wild-type PRRSV-2 RNA by RT-qPCR and open reading frame 5 (ORF)− 5 sequencing. Regression models were used to compare wild-type PRRSV detection dynamics on oral fluids samples and to compare key closeout performance indicators between one dose group and two doses group. Additionally, a benefit-cost ratio analysis compared economic performance between one dose group and two doses group. The proportion of wild-type PRRSV detection on oral fluids samples and the log counts of viral RNA per ml of oral fluids from the two doses group was lower than the one dose group on lots originated from PRRSV positive-stable sow farms, with a risk ratio of 1.24 and a rate ratio of 1.17, respectively. The two doses group had a significantly lower mortality rate than the one dose group, with a rate ratio of 1.21. The effect size increased on lots originated from PRRSV positive-unstable sow farms, and on lots with higher frequency and diversity of wild-type PRRSV detection during the growth phase. No differences in growth performance were detected between two doses group and one dose group. The second MLV vaccination dose had a benefit-cost ratio of 1.83. For lots originated from PRRSV positive-unstable farms, the benefit-cost ratio was 4.45, and for lots originated from PRRSV positive-stable farms, the benefit-cost ratio was 0.45. Under study conditions, vaccinating growing pig lots with two doses of PRRS MLV vaccine was a useful strategy to immunize growing pigs against PRRSV, lowering the wild-type PRRSV detection, lowering mortality rate, and increasing profitability, compared to lots of growing pigs that received a single dose of the same vaccine.

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1. Introduction

Porcine reproductive and respiratory syndrome virus (PRRSV) infects pig populations, causing reproductive disorders and systemic disease (Zimmerman et al., 2019). The agent has been present in the global swine industry for over 30 years and still causes significant economic impact to swine production (Neumann et al., 2005; Holtkamp et al., 2013). Holtkamp et al. (2013) estimated that 55% of this impact is on the growing pig phase (Holtkamp et al., 2013).

Over the last years, the swine industry has been developing strategies to mitigate the consequences of virus infection. Such practices consist of creating and improving population-based methods for PRRSV diagnostics in swine (Rotolo et al., 2017; Kittawornrat et al., 2012; Prickett et al., 2008; Wernike et al., 2012; Almeida et al., 2019; Lopez et al., 2018; Trevisan et al., 2019; Trevisan et al., 2019; Ramirez et al., 2012) and developing cost-effective immunization strategies for sows and pigs (Moura et al., 2019; Linhares et al., 2015, 2014; Corzo et al., 2010; Pileri and Mateu, 2016; Murtaugh and Genzow, 2011).

It has been demonstrated that PRRS modified-live virus (MLV) vaccination with one full dose significantly reduces clinical consequences of PRRSV challenge compared to non-vaccinated pigs (Linhares et al., 2012; Zuckermann et al., 2007; Cano et al., 2016; Lyoo et al., 2016; Kristensen et al., 2018; Madapong et al., 2017; Jeong et al., 2016). Aiming to further improve the protection against heterologous field viruses, researchers worldwide have developed new PRRSV immunization strategies (Bernelin-Cottet et al., 2019; Cao et al., 2018; Nan et al., 2017; Sirisereewan et al., 2017; Binjawadagi et al., 2016; Murthy et al., 2015; Peng et al., 2013; Li et al., 2013; Dwivedi et al., 2011).

In a case study from 1998, the application of two PRRS MLV vaccine doses, thirty days apart, effectively reduced the spread of wild-type virus in growing pig populations (Dee and Philips, 1998). However, there is limited information on how this strategy of vaccinating growing pigs with two PRRS MLV vaccine doses improves the protection against field viruses compared to vaccination with a single PRRSV MLV vaccine dose.

The objectives of this study were: (1) compare the frequency of wild-type PRRSV detection in oral fluids during the wean-to-finish phase, (2) compare key closeout production indicators, and (3) compare the economic performance between lots of growing pigs vaccinated with two doses of Ingelvac PRRS® MLV vaccine and lots vaccinated with a single dose of the same vaccine. We hypothesized that the second dose of PRRSV MLV vaccine would significantly reduce wild-type viral detection and improve the productivity performance of growing pig populations subsequently naturally challenged with wild-type PRRSV.

2. Materials and methods

2.1. Study design

This was a multi-source, randomized blocked field trial, conducted from the Fall of 2017 to the Summer of 2019, to compare virus detection, productivity, and economic performance of growing pig populations immunized with two different protocols of Ingelvac PRRS® MLV vaccination, in commercial swine operations of USA (Fig. 1). Nine PRRSV vaccinated sow farms (from three production systems representing the contemporary industry) that followed a specific eligibility criteria were conveniently selected to participate and their subsequent growing pig lots were included in the study. The production system A included 1 PRRSV positive-stable sow farm, the production system B included 2 PRRSV positive-unstable sow farms and 4 PRRSV positive-stable sow farms, and the production system C included 2 PRRSV positive-stable sow farms. A growing pig lot was defined as cohort of pigs placed in all-in all-out growing pig operations from wean to market. A randomized complete blocking design was used to allocate the two vaccination protocols to the lots of growing pigs. One block with sixteen lots of growing pigs from each sow farm of the production system A was created, while for production systems B and C, the blocks were created for every two consecutive lots of growing pigs from each sow farm. A fair coin was tossed to determine the vaccination protocol of the first half of the lots in each block. The second half of the lots automatically received the other vaccination protocol.

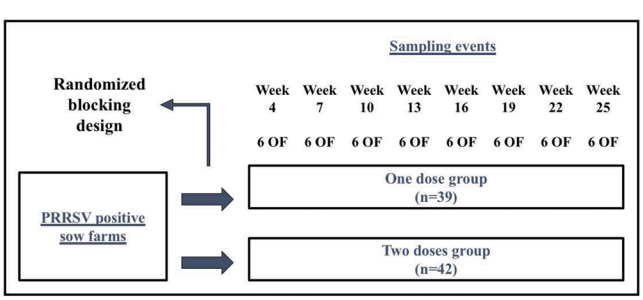

Fig. 1. Overview of the study design.

All the randomization and blocking process was done by the principal investigator of the study (CAAM) before the study began. Blinding of those administering the interventions and the caregivers was not possible, as the research was done in field conditions, and caregivers were the ones administering the vaccination events. The Iowa State University institutional animal care and use committee (IACUC) reviewed and approved this study protocol on June 15th of 2017 (log number 6–17–8538-S).

2.2. Eligibility and exclusion criteria of the study population

All lots of growing pigs included in the study were selected based on the following eligibility criteria: (1) all pigs in the lot originated from PRRSV positive-unstable sow farms (farm producing PRRSV viremic piglets at weaning) or PRRSV positive-stable sow farms (farm consistently producing PRRSV non-viremic piglets at weaning) according to the 2011 American Association of Swine Veterinarians (AASV) PRRSV classification (Holtkamp et al., 2011); (2) pigs from sow farms of different PRRSV status were not commingled; (3) placed in high pig dense areas, defined as the region having more than 185 pigs/km^2 (Alkhamis et al., 2018), (4) housed in all-in-all-out flows by site, and (5) group integrity maintained from weaning until marketing. Additionally, eligibility criteria for production systems included (1) willingness to participate, (2) ability to provide eligible lots of growing pigs and (3) availability to provide farm staff to collect and ship all the oral fluids according to the study protocol.

The exclusion criteria of lots of growing pigs included (1) reports of non-PRRS disease outbreaks of agents that impact the closeout performance indicators of experimental units, including porcine epidemic diarrhea virus (PEDV) and *Actinobacillus pleuropneumoniae* (App) outbreaks, and (2) report of non-infectious events or conditions that impact closeout performance indicators of pig lots.

2.3. Vaccination program

All 81 lots of growing pigs received the first dose of 2 ml of Ingelvac PRRS® MLV vaccine (Boehringer Ingelheim Animal Health USA Inc., Duluth, GA.) via the intramuscular route, either around processing age (3–5 days of age) or around weaning age (19–21 days of age). Lots allocated in the two doses group received a second dose of 2 ml of Ingelvac PRRS® MLV vaccine, via intramuscular route, approximately three to four weeks from the first vaccination.

Lots of growing pigs from the production systems A and C received the first vaccine dose at weaning age, while lots of growing pigs from production system B received the first vaccine dose at processing age. The blocking design matched lots of pigs from one dose and two doses groups by same sow farms and by the timing of the first vaccination dose.

2.4. Diagnostic monitoring

All 81 lots of growing pigs included in the analysis were monitored for wild-type PRRSV detection during the wean-to-finish phase by oral fluids. Six cotton ropes were spatially distributed in each lot of growing pigs in three weeks intervals, totaling eight sampling events between weaning and marketing. All samples were sent to Iowa State University Veterinary Diagnostic Laboratory and tested individually to detect PRRSV-2 RNA by RT-qPCR (quantitative reverse transcription polymerase chain reaction), which reported Cycle threshold (Ct) values, and PRRSV RNA copies/ml. If at least one of the six oral fluids samples were RT-qPCR positive, the sample with the lowest Ct value was submitted for PRRSV open reading frame 5 (ORF-5) genetic sequencing. Sampling events with ORF-5 sequencing results reporting more than 1% of genetic differences from the ORF-5 of the Ingelvac PRRS® MLV vaccine were defined as wild-type PRRSV (same criterion of Iowa State University Veterinary Diagnostic Laboratory). The RFLP (restriction fragment length polymorphism) cut pattern of each sampling event was also collected from the diagnostic report. The frequency of wild-type PRRSV detection was measured by the number of positive wild-type PRRSV sampling events at each growing pig lot. The diversity of wild-type PRRSV detection was measured by the number of unique RFLPs detected at each growing pig lot.

2.5. Statistical analyses

Statistical analyses were performed using each lot of growing pigs (populations) as the experimental unit. The sample size was calculated to detect at least a 0.5% difference (4.5–4%) on mortality rate between the 2 study groups (one dose vs. two dose groups). The sample required was 53,249 per treatment group, assuming a Poisson distribution, with a power of 80%, 95% confidence level and ratio between groups of 1. In order to achieve the sample size, 81 groups containing at least 660 growing pigs were included in the study.

Regression models were used to compare wild-type PRRSV RNA detection on oral fluids samples between two doses and one dose groups. Binomial regression models were built on SAS 9.4 (SAS Institute, Inc., Cary, NC) to model the proportion of wild-type PRRSV detection on oral fluids samples at each sampling event. The proportion of wild-type PRRSV detection was calculated by dividing the number of PCR-positive oral fluids samples by six, only for sampling events defined as wild-type PRRSV. P-values and risk ratios with 95% confidence intervals were reported. Poisson regression models were built on SAS 9.4 to model the log counts of viral RNA per ml of oral fluids at each sampling event. The counts of genetic copies of PRRSV RNA per ml of oral fluids was assessed only for oral fluids samples from sampling events defined as wild-type PRRSV. The log counts of viral RNA per ml of oral fluids at each sampling event was calculated by log (sum of the counts of genetic copies of PRRSV RNA per ml of oral fluids of all positive samples + 1). P-values and rate ratios with 95% confidence intervals were reported.

Generalized linear mixed regression models were used to compare key closeout productivity indicators between two doses and one dose groups. Poisson regression models were built on SAS 9.4 to model the mortality rate of each growing pig lot, by the counts of dead pigs adjusted by the number of pigs started as an offset variable. The P-values and the rate ratios with 95% confidence intervals were reported. Linear mixed regression models were built on SAS 9.4 to model average daily gain and feed conversion of each growing pig lot. P-values and mean differences with 95% confidence intervals were reported. PRRSV status of the sow farm and timing of the vaccination were used as fixed effects in the models. Region, system, sow farm, and blocks were used as random effects in the models. All statistical analyses were done using a significance level of 0.05.

A benefit-cost ratio analysis was used to compare economic performance between two doses and one dose groups. A production-economic model was built on Microsoft Excel (version 16.16.26, Microsoft Corporation, Redmond, WA) to estimate the impact that the changes in the productivity parameters described by the regression models have in economic terms. The model calculated the economic benefit of adding the second MLV vaccination dose to lots of growing pigs by the ratio of the difference in net profit over the difference in vaccination costs, between two doses and one dose groups.

The model used the least square means reported by the regression models to estimate the differences in productivity between lots of 10,000 growing pigs from one dose and two doses groups. The numerical differences were only included if the significance level was lower than 0.25 in the regression models. The revenue by group was calculated based on the total weight of market pig produced, with market pig prices of $1.14 per kg ($140,00 for a 122 kg pig). Production costs by group were calculated based on fixed costs of $11.79/pig, weaned pig costs of $42.86/pig, diet costs of $0.24/kg of feed, and variable costs of $19.75/pig (Table 1). The prices applied to this study were calculated based on the monthly average sales value in dollars from December of 2018 to November of 2019 (http://www2.econ.iastate.edu/estimated-returns). The net profit of one dose and two doses groups were calculated based on the difference of revenue and production costs. The vaccination costs of one dose and two doses groups were calculated based on the price of $1,00 per vaccine dose and $0.10 per vaccination labor. The ratio of the difference in net profit over the difference in vaccination costs between two doses and one dose groups was reported as a benefit-cost ratio. The second MLV vaccination dose was considered economically beneficial if the benefit-cost ratio was greater than 1.

A sensitivity analysis was conducted using Data Table function on Microsoft Excel to describe changes in the benefit-cost ratio of the second MLV vaccination dose, with varying vaccination costs and with different market pig prices. Price changes applied in sensitivity analysis were based on changes of $20 on the value of the market pig and $0.20 on the cost of the second vaccination dose.

3. Results

The study included 42 lots of growing pigs vaccinated with two doses and 39 lots vaccinated with one dose of Ingelvac PRRS® MLV vaccine (Table 2). There were 136,482 pigs, 634 sample collection events, and 3646 oral fluid samples. Lots of growing pigs of the same production system followed the same management practices, were housed at the same pig densities, were fed the same diets, and received the same water treatments during study period.

3.1. Wild-type virus detection

For lots originated from PRRSV positive-stable sow farms, the proportion of wild-type PRRSV detection on oral fluids samples from the

Table 1
Productivity and economic parameters of one dose and two doses groups.

	One dose group	Two doses group
Productivity		
Pigs placed (pigs/group)	10,000	10,000
Average days on feed (days/pigs)	160	160
Average start weight (kg/pig)	5.44	5.44
Wean to finish mortality (% of pigs placed)	10.54%	8.68%
Wean to finish ADG (kg/day)	0.7805	0.7805
Wean to finish FC (kg of feed/kg of gain)	2.2938	2.2749
Revenue and production costs		
Market pig price ($/kg)	$1.14	$1.14
Fixed costs ($/pig)	$11.79	$11.79
Weaned pig costs ($/pig)	$42.86	$42.86
Feed costs ($/kg of feed)	$0.24	$0.24
Other variable costs ($/pig)	$19.75	$19.75
Vaccination costs		
Vaccine cost ($/pig)	$1.00	2.00
Vaccination cost ($/pig)	$0.10	$0.20

Table 2
Demographic characteristics of one dose and two doses groups.

Sow farm PRRSV status	PRRSV positive-stable (n = 7)		PRRSV positive-unstable (n = 2)	
Study group	One dose group	Two doses group	One dose group	Two doses group
Number of lots included	33	33	6	9
Number of pigs included (mean ± standard deviation)	61,108 ± 1303	59,055 ± 1087	7403 ± 369	8916 ± 262
Weight started, kg (mean ± standard deviation)	6.75 ± 0.94	6.76 ± 0.95	6.16 ± 0.02	6.19 ± 0.12
Average days on feed (mean ± standard deviation)	161 ± 7.56	160 ± 7.01	154 ± 2.97	155 ± 5.75

* There was no statistical difference of any characteristic between groups at $P < 0.05$.

one dose group (21.23%) was significantly higher than the proportion from the two doses group (17.06%) ($p = 0.0082$) with a risk ratio of 1.24 (95% CI: 1.07 – 1.43). However, when lots were originated from PRRSV positive-unstable sow farms, the proportion of wild-type PRRSV detection on oral fluids samples between one dose group (53.65%) and two doses group (49.92%) was not significantly different ($p = 0.3707$). The comparisons of the proportion of wild-type PRRSV detection in oral fluids between one dose and two doses groups at each of the sampling events are shown in Fig. 2.

For lots originated from PRRSV positive-stable sow farms, the log counts of viral RNA per ml of oral fluids from the one dose group (1.29 copies/ml) were not significantly higher than that of the two doses group (1.10 copies/ml) ($p = 0.0532$), with a rate ratio of 1.17 (95% CI: 0.99 – 1.37). For lots originated from PRRSV positive-unstable sow farms, the log counts of viral RNA per ml of oral fluids between one dose group (2.74 copies/ml) and two doses group (2.53 copies/ml) were not significantly different ($p = 0.4848$). The comparisons of the log counts of viral RNA per ml of oral fluids between one dose and two doses groups at each of the sampling events are shown in Fig. 3.

3.2. Productivity performance

Overall, the one dose group had 10.54% of wean-to-finish mortality, and the two doses group had 8.68%. The p-value was < 0.0001 and the risk ratio was 1.21 (95% CI: 1.16–1.27). Considering only lots originated from PRRSV positive-unstable herds, the one dose group had 17.99% of wean-to-finish mortality, and the two doses group had 13.52%. This represents a reduction of 4.47% points on the wean-to-finish mortality of the two doses group compared to the one dose group, with a p-value < 0.0001 and risk ratio of 1.30 (95% CI: 1.22–1.44). All lots of growing pigs originated from PRRSV positive-unstable sow farms received the first vaccination at processing age. Considering only lots originated from PRRSV positive-stable herds, the one dose group had 6.18% of wean-to-finish mortality, and the two doses group had 5.57%. This represents a reduction of 0.61% points on the wean-to-finish mortality of the two doses group compared to the one dose group, with a p-value of 0.0001 and risk ratio of 1.11 (95% CI: 1.05–1.16) (Fig. 4). Lots of growing pigs originated from PRRSV positive-stable sow farms received the first vaccination either at processing or at weaning. The difference in wean-to-finish mortality between study groups was not significantly different across lots vaccinated with the first dose at processing and lots vaccinated with the first dose at weaning (p-value=0.2855).

The effect of the second MLV vaccination dose to reduce wean-to-finish mortality increased when there was higher frequency of wild-type PRRSV detection and higher diversity of wild-type detection during the growth phase. The mortality rate of one dose group was lower than that of two doses group when there was no wild-type PRRSV detection during the growth phase (Fig. 5). No significant differences in average daily gain (Table 3) and feed conversion (Table 4) were detected between one dose and two doses groups.

3.3. Economic performance

Based on the changes in productivity reported by the regression models, the net profit of the two doses group was $20,242 higher than the net profit of the one dose group. The vaccination costs with two doses was $11,000 higher than the costs of vaccinating pigs with only one dose. As the difference in net profit was higher than the difference in vaccination costs, the second MLV vaccination dose was considered economically beneficial, with a benefit-cost ratio of 1.83 (Table 5). When including only lots from PRRSV positive-unstable sow farms, the benefit cost-ratio of the second MLV vaccination was 4.45, while for lots originated from PRRSV positive-stable sow farms, the benefit cost-ratio of the second MLV vaccination was 0.45.

The second MLV vaccination dose was economically beneficial when vaccination costs were lower than $2.17 per dose or when the market pig prices were higher than $0.74/kg. Differences in the benefit-cost ratio of the second MLV vaccination with varying market pig prices and varying vaccination costs are described in Fig. 6.

4. Discussion

This multi-source randomized field study sought to characterize

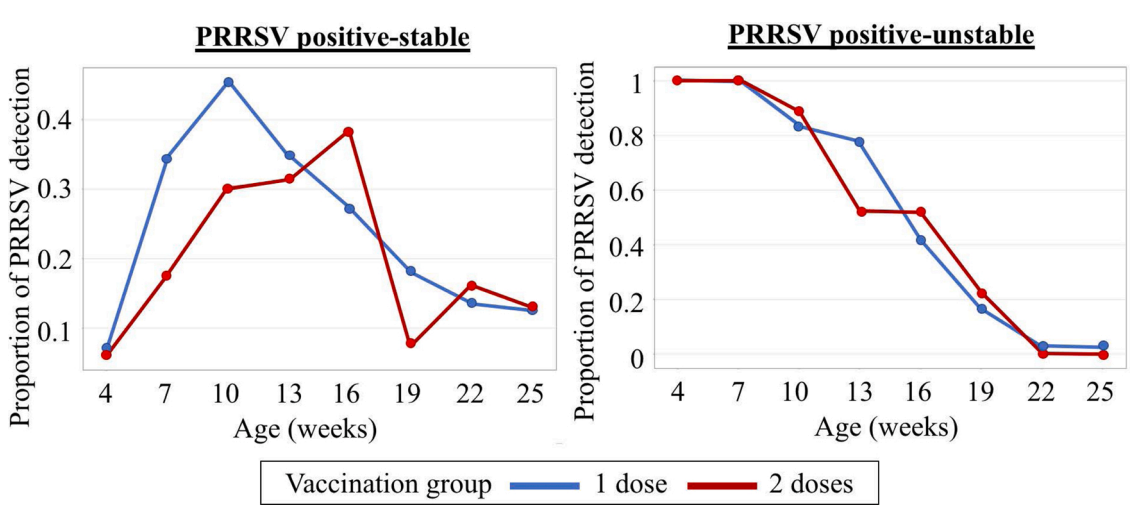

Fig. 2. Proportion of wild-type PRRSV detection in oral fluids samples over time for one dose and two doses groups.

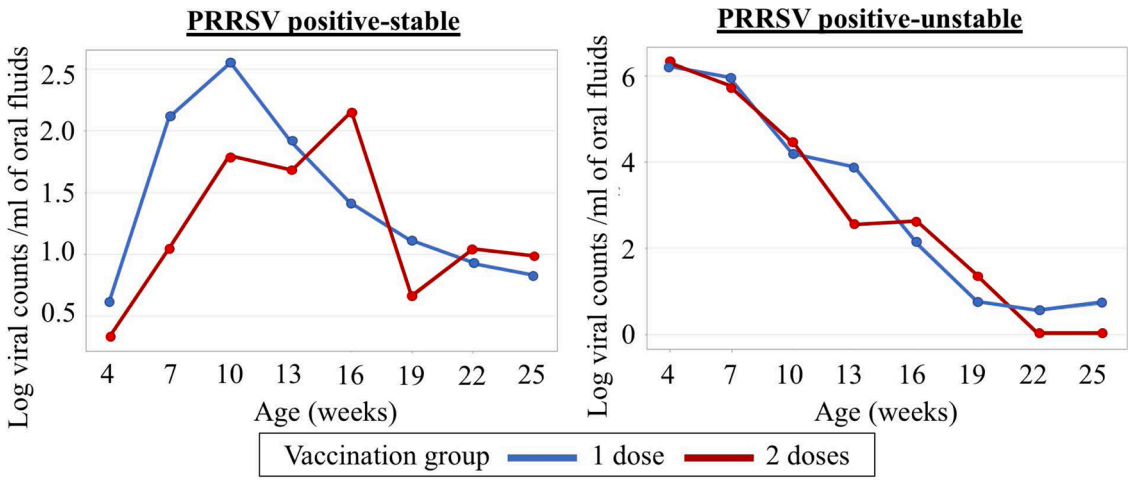

Fig. 3. Log counts of wild-type PRRSV RNA per ml of oral fluids over time for one dose and two doses groups.

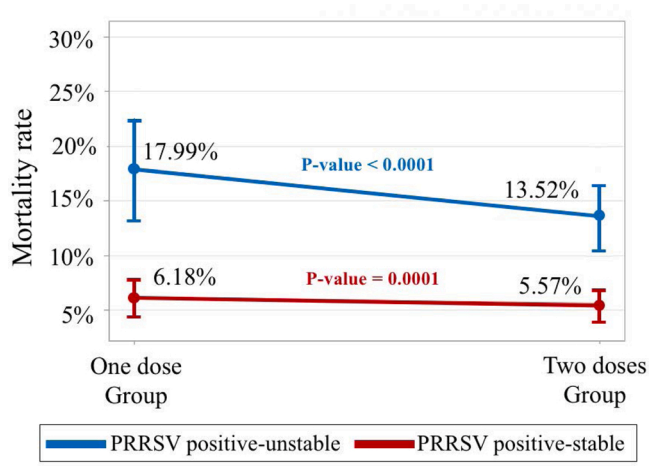

Fig. 4. Comparison of wean-to-finish mortality rate between one dose and two doses groups, by source sow farm PRRSV status (means and standard errors).

wild-type PRRSV detection on oral fluids, as well as productivity and economic performance of lots of growing pigs immunized with two doses of Ingelvac PRRS® MLV vaccine, compared to lots vaccinated with a single dose of the same vaccine. A randomized blocking design was used to allocate the two vaccination protocols into the lots of pigs. The growing pig populations of this study were selected to reflect those originated from PRRSV-positive breeding herds, and placed in high-pig dense areas (i.e., high risk of exposure to wild-type PRRSV). These criteria were implemented to assure natural wild-type PRRSV exposure to the growing pig lots since this was a field trial, and no experimental PRRSV challenge model was implemented.

Overall, the wild-type PRRSV detection in oral fluids samples from two doses group was lower than that of one dose group. A reduction in the extent of wild-type PRRSV detection on oral fluids samples from two

Table 3
Wean-to-finish average daily gain (ADG) comparison between one dose and two doses groups.

Status	One dose group ADG	Two doses group ADG	ADG difference (95% CI)	Sample size	p-value
Overall	0.7805	0.7853	-0.005(−0.040; 0.037)	81	0.7387
Positive-unstable	0.7813	0.7775	0.004(−0.063; 0.069)	15	0.9132
Positive-stable	0.7840	0.7906	-0.007(−0.039; 0.026)	66	0.6786

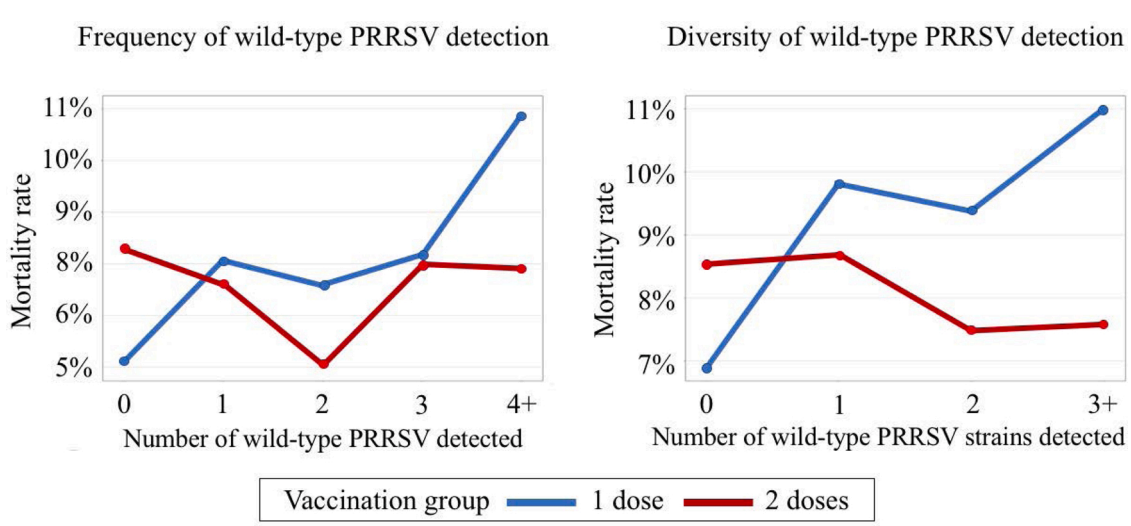

Fig. 5. Comparison of wean-to-finish mortality rate between one dose and two doses groups, by the frequency and the diversity of wild-type PRRSV detection.

Table 4
Wean-to-finish feed conversion (FC) comparison between one dose and two doses groups.

Status	One dose group FC	Two doses group FC	FC Difference (95% CI)	Sample size	p-value
Overall	2.2938	2.2749	0.0189 (−0.012; 0.049)	81	0.2212
PRRSV Positive-unstable	2.3110	2.2600	0.0510 (−0.022; 0.123)	15	0.1679
PRRSV Positive-stable	2.2923	2.2802	0.012 (−0.022; 0.046)	66	0.4769

Table 5
Economic performance comparison between one dose and two doses groups.

Economic parameters	One dose group	Two doses group	Difference
Productivity			
Average final weight (kg/pig)	130.3	130.3	0
Total weight produced (kg/group)	1,165,843	1,190,082	24,240
Feed fed per group (kg of feed/group)	2,562,579	2,594,305	31,726
Revenue and production costs			
Revenue ($/group)	$1,332,719	$1,360,428	$27,709
Fixed costs ($/group)	$117,900	$117,900	$0
Weaned pig costs ($/group)	$428,600	$428,600	$0
Feed costs ($/group)	$603,139	$610,606	$7467
Other variable costs ($/group)	$197,500	$197,500	$0
Net profit ($/group)	-$14,421	$5821	**$20,242**
Vaccination costs			
Vaccine and labor costs ($/group)	$11,000	$22,000	**$11,000**
Benefit-cost ratio			**1.83**

		Market pig price ($/kg)				
		$0.82	**$0.98**	**$1.14**	**$1.31**	**$1.47**
Vaccination cost (vaccine + labor)	**$1.50**	0.83	1.09	1.34	1.62	1.88
	$1.30	0.95	1.25	1.55	1.87	2.17
	$1.10	1.13	1.48	1.83	2.21	2.56
	$0.90	1.38	1.81	2.24	2.70	3.13
	$0.70	1.77	2.33	2.88	3.47	4.02

Fig. 6. Sensitivity analysis of the benefit-cost ratio varying vaccination costs and market pig prices.

doses group, compared to the one dose group, was more evident on lots from PRRSV positive-stable sow farms than in lots originated from PRRSV positive-unstable sow farms. The effect size of the reduction in detection of two doses group from PRRSV positive-stable sow farms, compared to one dose group, was higher on the sampling events collected at weeks seven, 10, and 19 of age. These results suggest a potential benefit for vaccinating pigs from PRRSV positive-stable sow farms with the two doses of PRRS MLV vaccine to reduce wild-type PRRSV detection during the wean-to-finish phase. Lower wild-type PRRSV detection at growing pig sites may be a proxy of lower virus circulation, which can help to reduce the risk of PRRSV outbreaks in neighbor sow farms (Trevisan et al., 2020). No major differences were detected between one dose and two doses groups originated from PRRSV positive-unstable sow farms, perhaps due to the high prevalence of wild-type PRRSV at the time of placement of pigs into the growing pig sites.

Differences in the RFLP cut patterns detected at each sampling event were used as a measure for the diversity of virus detection within each lot of growing pigs. In the absence of other practical methods, the RFLP was selected in order to facilitate communication across field veterinarians, since it is reported in diagnostic report from most swine-centric veterinary diagnostic laboratories in the USA. Results describing the diversity of virus detection over time using the RFLP cut patterns were similar to those obtained when using a cut off of 1% of genetic differences in the ORF-5 sequencing (data not presented).

Significant differences (at alpha level of 0.05) in productivity parameters between two doses and one dose groups were detected only for mortality. The effect size was higher on lots originated from PRRSV positive-unstable sow farms than in lots from PRRSV positive-stable sow farms. The effect size also increased in lots with a higher frequency of wild-type PRRSV detection and with higher diversity of wild-type PRRSV detection during the growing phase. The Poisson regression model was very sensitive for detecting differences in mortality between vaccination groups, however significant differences in growth performance parameters were not detected by the linear regression models. The sample size obtained in this study only had 5% of power to detect statistical significance in the observed average daily gain difference and 16% of power to detect statistical significance in the observed feed conversion difference between one dose and two doses groups. In order to detect significant differences with 80% of power, the differences in average daily gain and in the feed conversion between one dose and two doses groups would have to be at least 0.047 and 0.055, respectively. Moreover, it is important to note that although no significant differences were detected on growth parameters, all measures of effects detected were reporting improvements on the two doses group.

While results from this study showed a reduction in mortality of lots of pigs vaccinated with two PRRSV MLV doses, recent studies proposing a DNA-MLV prime-boost strategy regimen (vaccinating pigs with a DNA vaccine and applying a PRRS MLV vaccine a few weeks afterwards) improved the immune response, but not the clinical response against heterologous PRRSV infection. The DNA-MLV prime-boost strategy improved PRRSV-specific cellular response and increased the neutralizing antibody titers on one study (Sirisereewan et al., 2017) and slightly improved the T cell response, with a substantial increase in the PRRSV specific antibody response on a different study (Bernelin-Cottet et al., 2019). In fact, the strategy of using two doses of PRRSV MLV vaccination on growing pigs from a unidirectional flow was effective to control PRRSV from finishing populations. The vaccination protocol was carried out with all pigs within the facility receiving an intramuscular vaccination with 2.0 ml of PRRS MLV vaccine, which was repeated thirty days later. Authors hypothesized that multiple doses of vaccine would help to reduce the spread of virus, and to reduce the risk of improperly vaccinating or nonvaccinating pigs existing in the population (Dee and Philips, 1998). Differences in productivity and in wild-type PRRSV detection between one dose and two doses groups might be due to improvements on the cell-mediated immune response (Savard et al., 2016), humoral immune response (Jeong et al., 2016; Bernelin-Cottet et al., 2019), or both (Jeong et al., 2016; Nan et al., 2017). However, further studies are necessary to clarify this research question.

The economic analysis presented in this study demonstrated that it was economically beneficial to use the second MLV vaccination dose to immunize lots of growing pigs. Based on the changes in productivity reported by the regression models, the improvement in the net profit of the two doses group represented 183% of the increase in the cost of vaccination. The second MLV vaccination was even more economically beneficial for lots originated from PRRSV positive-unstable sow farms, due to the higher differences in productivity reported. However, for lots originated from PRRSV positive-stable sow farms, the second MLV dose was only economically beneficial when market pig prices were higher than $1.47/kg and when vaccination costs were lower than $0.75. The prices applied to this study were calculated based on the monthly average sales value in dollars from December of 2018 to November of 2019 (http://www2.econ.iastate.edu/estimated-returns).

The vaccine applications were made by farm staff, and no adverse effects were reported for neither one dose nor the two doses groups. A

limitation of this study was that it did not include diagnostic information of other pathogens of interest in swine production, nor information on causes of death of pigs. Another limitation was that the study did not include a non-vaccinated control group. Also, it is important to note that there is expected variation of PRRSV impact on productivity as time progresses from the outbreak. Thus, some variation in growth performance may have been attributed to timing from the outreak for the Positive Unstable groups. The process taken to randomly allocate the treatments, and the fact that treatment groups were matched by source and blocked by timing from outbreak accounted for that variation.

5. Conclusions

The proportion of wild-type PRRSV detection on oral fluids samples from the two doses group was lower than the one dose group when lots originated from PRRSV positive-stable sow farms, with a risk ratio of 1.24. The two doses group had a significantly lower mortality rate than the one dose group, with a rate ratio of 1.21. The effect size was higher on lots from PRRSV positive-unstable sow farms, and on lots with higher frequency and diversity of wild-type PRRSV detection during the growing pig phase. No differences in growth performance (average daily gain or feed efficiency) were detected between two doses and one dose groups. The second MLV vaccination dose was economically beneficial, with benefit-cost ratio of 1.83. For lots originated from PRRSV positive-unstable farms, the benefit-cost ratio was 4.45 and for lots originated from PRRSV positive-stable farms, the benefit-cost ratio was 0.45.

Under the conditions of this study, vaccinating growing pig lots with two full doses of PRRS MLV vaccine was a useful strategy to immunize growing pigs against PRRSV, lowering the wild-type PRRSV detection, lowering growing pig mortality rate, and increasing growing pig profitability, compared to lots of growing pigs that received a single dose of the same vaccine.

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Declaration of Competing Interest

RP is employed by Boehringer Ingelheim Animal Health.

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